



Lab 04

Fixed-Bias and Emitter-Bias BJT Biasing Configurations

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4.1 Objective

To investigate the effect of current gain β variation on quiescent operating conditions of a Bipolar Junction Transistor in fixed bias and emitter bias configurations.

Note: All the plots required to be attached should be labeled properly, followed by Task number and must be readable, i.e., use white background, suitable marker color (recommended – any dark color) and increase the width of trace. On PSpice there is an option to add text labels in the plots.
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4.2 Introduction to BJT Biasing

The term biasing refers to the application of DC voltages at transistor terminals to establish a fixed level of current and voltage. The operating point (I_C , V_{CE}) is a fixed point on the characteristics, it is also called the quiescent point (abbreviated Q -point). By definition, quiescent means quiet, still, inactive. The chosen Q -point often depends on the intended use of the circuit like application of transistor as amplifier or switch.

Since, temperature causes the device parameters such as the transistor current gain and the transistor leakage current (I_{CEO}) to change. Higher temperatures result in increased leakage currents in the device, thereby changing the operating condition set by the biasing network. Therefore, the network design must also provide a degree of temperature stability so that temperature changes result in minimum changes in the operating point. This maintenance of the operating point can be specified by a stability factor S , which indicates the degree of change in operating point due to variation in parameters. A highly stable circuit is desirable. In this lab, two biasing circuits will be investigated and their stability will be compared. After that, you will be asked to design a biasing network for given operating point.



4.3 Fixed Bias Configuration

The fixed-bias circuit of Figure is the simplest transistor DC bias configuration.

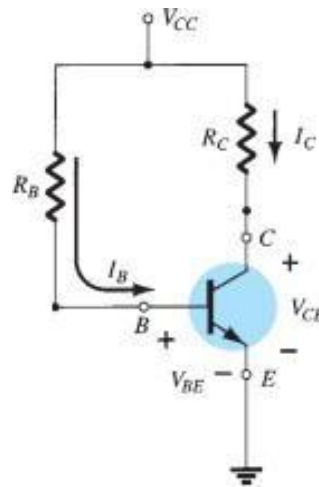


Figure 4.1: Fixed bias BJT configuration

Analysis of the circuit can be done by considering its two loops; the base-emitter loop and the collector-emitter loop. From base-emitter loop, the base current I_B is given by;

$$I_B = \frac{V_{CC} - V_{BE}}{R_B}$$

The magnitude of collector current I_C is related to the base current I_B through collector-to-emitter voltage V_{CE} is given by

$$V_{CE} = V_{CC} - I_C R_C$$

Note that here, V_E is 0 V as emitter is directly grounded.

Task 1: To calculate the operating point of BJT in given fixed-bias configuration

Consider a BJT 2N3904 biased in fixed-bias configuration shown in Figure 4.1 with V_{CC} being 20V, R_B of 1 M Ω and R_C of 2.7 k Ω . Using the given formulae, perform calculations to find the operating point of BJT. Use $\beta = 140$ and 0.7 V base-to-emitter forward voltage for the given transistor in your calculations.

Table 4.1: Calculated values of operating point of 2N3904 in given fixed bias configuration

Parameter	Calculated Value
Base-to-Emitter Voltage: V_{BE}	0.7 V (Given)
Current Gain: β	140 (Given)
Base Current: I_B	1.93×10^{-5} A



Collector Current: I_C	2.702 mA
Collector-to-Emitter Voltage: V_{CE}	12.7046 V



Task 2: To implement fixed-bias configuration circuit with 2N3904 and 2N4401 BJT.

1. Construct the circuit of Figure 4.1 in with the component values as given in Task 1 for 2N3904 BJT model.
2. Do not use a multimeter to identify terminals. Use the datasheet of 2N3904 for its pin configuration.
3. Measure the parameters listed in Table 4.2 to find the operating point of BJT.
4. Necessary precautions shall be ensured while using DMM Ammeter to avoid damage to the equipment. After measuring the current the probe is to be always put back to voltage section.

Table 4.2: Measured results of fixed-bias configuration with 2 different BJT models

Parameter	Measured Results	
	2N3904	2N4401
Base Current: I_B	19.96 μ A	19.97 μ A
Collector Current: I_C	6.3mA	5.5mA
Collector-to-Emitter Voltage: V_{CE}	3.36 V	5.27 V
Current Gain: β (calculated)	315.63	275.41
Base-to-Emitter Voltage: V_{BE}	0.684 V	0.674 V
Measured Resistances	$R_c = 2.68k$ ohms, $R_b = 0.97M$ ohms	$R_c = 2.68k$ ohms, $R_b = 0.97M$ ohms

5. Are the obtained results same as the calculated ones? What can be a possible reason behind discrepancies (if any)? Compare the value of current gain β .

The values are a bit different, especially for I_C , this is due to the internal inheritance of the transistor which differs the current gain (β). β is also different compared to what was given.

6. Since, the current gain β of a transistor model varies from unit to unit and with operating conditions of temperature, collector current, collector-to-emitter voltage etc. Therefore, to further investigate the effect of change in β on operating point of circuit, we will study the same circuit for a different BJT model 2N4401 (whose rated β as per datasheet is different from the previously used model 2N3904).
7. Modify the circuit by replacing the BJT by 2N4401 model and retake measurements to complete the corresponding column of Table 4.2.



Task 3: To investigate effect of current gain β variation on BJT operating point with fixed-bias configuration

1. To investigate the effect of variation in β on the operating point of transistors in fixed bias configuration, calculate the parameters given in Table 4.3 and fill in the Fixed-Bias Configuration column.

Table 4.3: Variation in BJT operating point w.r.t. the variation in DC current gain β

Percentage Change in Parameters	Fixed-Bias Configuration	Emitter-Bias Configuration
$\% \Delta \beta = \left \frac{\beta_{4401} - \beta_{3904}}{\beta_{3904}} \right \times 100\%$	12.74%	13.4%
$\% \Delta I_C = \left \frac{I_{C4401} - I_{C3904}}{I_{C3904}} \right \times 100\%$	12.70%	8.55%
$\% \Delta V_{CE} = \left \frac{V_{CE4401} - V_{CE3904}}{V_{CE3904}} \right \times 100\%$	56.85%	29.5%
$\% \Delta I_B = \left \frac{I_{B4401} - I_{B3904}}{I_{B3904}} \right \times 100\%$	0.05%	5.6%

2. Compare operating points of the two BJT models when biased using fixed-bias configuration. Analyze the effect of variation in β on each of the parameters one by one for fixed-bias configuration.

Change in I_C is equivalent to change in β , change in V_{ce} is much greater compared to change in β , it shows a small change in β causes I_C to change by same rate but V_{ce} increases by a greater percentage.

Change in I_B remains very small.



4.4 Emitter Bias Configuration

Emitter biasing circuit looks same as the fixed-bias configuration but it contains a resistor connected at emitter terminal called emitter resistor R_E . The effect of adding this resistor in the circuit will be investigated in the following tasks.

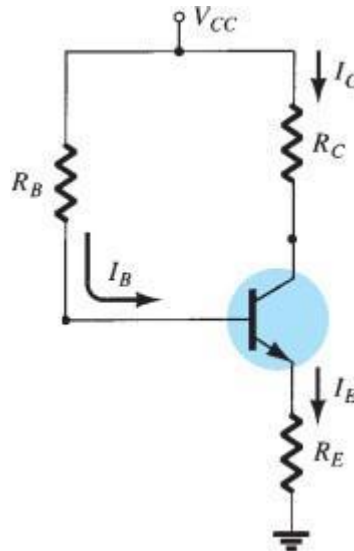


Figure 4.2: Emitter-biased configuration circuit

Here again, analysis of the circuit can be done by considering its two loops; the base-emitter loop and the collector-emitter loop. Applying KVL to the base-emitter loop

$$V_{CC} - I_B R_B - V_{BE} - I_E R_E = 0$$

Since, $I_E = (\beta + 1)I_B$. Therefore, the base current I_B is given by;

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E}$$

Note, that the equation for base current now includes $(\beta + 1)R_E$.

The emitter resistor R_E , which is part of the collector-emitter loop, is reflected back to the input base circuit by a factor $(\beta + 1)$.

As $I_C \approx I_E$, therefore, for collector-emitter loop the KVL equation will result in the following;

$$V_{CE} = V_{CC} - I_C(R_C + R_E)$$

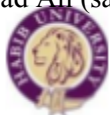
Note that for emitter-biased configuration, the emitter is not directly connected to ground and hence, emitter voltage with respect to ground is given as

$$V_E = I_E R_E$$

This can be used to calculate base voltage and collector voltage with respect to ground. The voltage from emitter to ground can't be unreasonably large and is typically around one-fourth to one-tenth of the supply voltage.

Task 4: To apply emitter-bias configuration for 2N3904 and 2N4401 BJT

1. Construct the circuit of Figure 4.2 with 2N3904 BJT model when V_{CC} is 20V, R_B is 1 M Ω , R_C is



2.2 k Ω and R_E is 2.2k Ω .

2. Measure the parameters listed in Table 4.4 to find the operating point of BJT.



3. Modify the circuit by replacing the BJT by 2N4401 model and retake measurements to complete the corresponding column of Table 4.4.

Table 4.4: Measured results of fixed-bias configuration with 2 different BJT models

Parameter	Measured Results	
	2N3904	2N4401
Base Current: I_B	11.72 μA	12.38 μA
Collector Current: I_C	3.718 mA	3.40 mA
Collector-to-Emitter Voltage: V_{CE}	4.03 V	5.22 V
Current Gain: β (calculated)	317.23	274.63
Base-to-Emitter Voltage: V_{BE}	0.665 V	0.66 V
Measured Resistances	$R_b = 0.97 \text{ M ohms}$, $R_c = 2.16 \text{ k ohms}$, $R_e = 2.156 \text{ k ohms}$	$R_b = 0.97 \text{ M ohms}$, $R_c = 2.16 \text{ k ohms}$, $R_e = 2.156 \text{ k ohms}$

Task 5: To investigate effect of current gain β variation on BJT operating point with emitter-bias configuration and compare stability of the 2 biasing techniques

1. Using the data collected in Table 4.4, calculate the percentage change in different parameters listed in Table 4.3 and fill in the respective column of Emitter-Bias Configuration.
2. Analyze the effect of variation in β on the operating point of BJT when biased through emitter-bias configuration.

Now, for almost same variation in β , both I_c and V_{ce} have relatively lower variation compared to fixed biases configurations. Whereas the change in I_b has increased

3. The stability of a system is a measure of the sensitivity of a network to variations in its parameters and one of these parameters is current gain β . A stability factor $S(\beta)$ is defined for the impact of β on bias stability as follows:

$$S(\beta) = \frac{\Delta I_c}{\Delta \beta}$$

$$\Delta \beta$$

Similarly, ratio of change in V_{CE} and change in β can be used together with the above given stability factor to check dependency of operating point on β variations.



4. Calculate the stability factors for each of the biasing techniques using Table 4.3 and fill in Table 4.5.

Table 4.5: Stability factor w.r.t. β variation for Fixed-Bias and Emitter-Bias configurations

Biasing Configuration	$\frac{\% \Delta I_C}{\% \Delta \beta}$	$\frac{\% \Delta V_{CE}}{\% \Delta \beta}$
Fixed-Bias	1	4.46
Emitter-Bias	0.638	2.2

5. From the stability factor and percentage change in parameters calculated for the two BJT biasing techniques, compare the configurations and justify which one of these is more stable i.e. lesser sensitive to the change in DC current gain.

Emitter bias is more suitable as we can see when β is varied, it has lesser variation in both I_C and V_C and when there is change in DC current gain compared to fixed bias making it more suitable and less sensitive to change in the conditions.

Task 6: To design biasing circuit for given operating point

1. For 2N3904 BJT, design an emitter-stabilized bias circuit to achieve an operating point such that collector current I_C is 2mA and collector-to-emitter voltage V_{CE} is 10V.

Use 20V supply voltage, DC current gain of 150 and select the conservative case of emitter voltage being one-tenth of the supply voltage in your design calculations and fill in Table 4.6. For resistors, you may approximate the calculated values as per the practically available resistors. Do not use the standard values for resistors, instead try to make them as close as possible.

Table 4.6: Design parameters of emitter-bias circuit for given operating point of BJT

Parameters	Calculated Values
Emitter Voltage: V_E	2 V
Emitter Resistor: R_E	1k ohms
Collector Resistor: R_C	4k ohms



Base Current: I_B	13.25 μA
Base Resistor: R_B	1.306M ohms

2. Construct your designed circuit and find out the operating point of BJT to complete Table 4.7.

Table 4.7: Comparison of BJT operating point for designed and implemented circuit

Operating Point	Estimated Q Point	Measured Q-Point
I_{CQ}	2 mA	1.997 mA
V_{CE_Q}	10 V	10.1 V

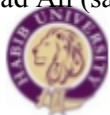
3. Compare the estimated Q-point for which the circuit was designed with the one obtained in measured results. Justify the analysis.

The estimated and the measured value is almost the same which tells us that our transistor is operation at an optimal current.

4.5 Post-Lab Task

Task 7: To list down the factors that affect BJT DC Current Gain

- What are the factors that affect the DC current gain β of a BJT?
 - Temperature, as when temperature increases β decreases
 - Collector current, as current increases, β decreases.
 - Physical property of transistor (BJT)





Assessment Rubric
Lab 04
Fixed-Bias and Emitter-Bias BJT Biasing Configurations

Name:	Student ID:
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Points Distribution

Task No.	LR1 Circuit Layout 18	LR 4 Data Collection 10	LR 6 Calculation 16	LR 10 Analysis 22	LR 11 Design 18	LR3 Trouble Shooting 12
Task 1	-	-	/4	-	-	
Task 2	/7	/5	-	/4	-	/4
Task 3	-	-	/4	/4	-	
Task 4	/7	/5	-	-	-	/4
Task 5	-	-	/8	/7	-	
Task 6	/4	-	-	-	18	/4
Task 7	-	-	-	/7	-	
	/18	/10	/16	/22	/18	/12
CLO Mapped	CLO 2	CLO 2	CLO 1	CLO 1	CLO 2	CLO2

Affective Domain Rubric		Points	CLO Mapped
AR 4	Report Submission	/4	CLO 4

CLO	Total Points	Points Obtained
1	38	
2	58	
4	4	
Total	100	

For description of different levels of the mapped rubrics, please refer the provided Lab Evaluation Assessment Rubrics and Affective Domain Assessment Rubrics.



Lab Evaluation Assessment Rubric

#	Assessment Elements	Level 1: Unsatisfactory Points 0-1	Level 2: Developing Points 2	Level 3: Good Points 3	Level 4: Exemplary Points 4
LR1	Circuit Layout	Connections between circuit components are mostly wrong. Circuit layout is cluttered. Needs guidance to make correct equipment/component connections.	Few of the connections made between circuit components are wrong. Circuit layout is not neat and clean as per standards.	Circuit layout and connections are correct but not all connections are neat and clean as per standards.	Neat, clean and correct connections of all the circuit components/equipment are made as per standard circuit diagram.
LR2	Program/Code/ Simulation Model/ Network Model	Program/code/simulation model/network model does not implement the required functionality and has several errors. The student is not able to utilize even the basic tools of the software.	Program/code/simulation model/network model has some errors and does not produce completely accurate results. Student has limited command on the basic tools of the software.	Program/code/simulation model/network model gives correct output but not efficiently implemented or implemented by computationally complex routine.	Program/code/simulation /network model is efficiently implemented and gives correct output. Student has full command on the basic tools of the software.
LR3	Troubleshooting	Unable to identify the fault/minimal effort show in troubleshooting.	Able to identify the fault but unable to remove it.	Able to identify the fault but partially removes it.	Able to identify the fault and takes necessary steps and actions to correct it.
LR4	Data Collection	Measurements are incomplete, inaccurate and imprecise. Observations are incomplete or not included. Symbols, units and significant figures are not included.	Measurements are somewhat inaccurate and imprecise. Observations are incomplete or vague. Major errors are there in using symbols, units and significant digits.	Measurements are mostly accurate. Observations are generally complete. Minor errors are present in using symbols, units and significant digits.	Measurements are both accurate and precise. Data collection is systematic. Observations are very thorough and include appropriate symbols, units and significant digits and task completed in due time.
LR5	Results & Plots	Figures/ graphs / tables are not developed or are poorly constructed with erroneous results. Titles, captions, units are not mentioned. Data is presented in an obscure manner.	Figures, graphs and tables are drawn but contain errors. Titles, captions, units are not accurate. Data presentation is not too clear.	All figures, graphs, tables are correctly drawn but contain minor errors or some of the details are missing.	Figures / graphs / tables are correctly drawn and appropriate titles/captions and proper units are mentioned. Data presentation is systematic.
LR6	Calculations	Formulae used and/or calculations are incorrect. Units are not mentioned with results.	Formulae used are correct but there are few mistakes in calculation which lead to incorrect results.	Formulae used and end results are correct. Complete working steps are not shown. Proper units are not mentioned with results.	Formulae used, end results and all calculations are correct with all the intermediate steps clearly shown. Units are mentioned with results.
LR10	Analysis	Results are not interpreted or are analyzed incorrectly.	Analysis of the obtained results is incomplete. Conclusions are not drawn.	Results are analyzed and concluded but are not well explained and justification is not provided with the help of reasoning.	Results are well analyzed supported with reasons. Good comparison is made between the theoretical and experimental results with correct and useful conclusion.
LR11	Design	Proposed design is unsubstantiated or does not satisfy most of given constraints. The breakdown of system into features is incomplete and still at lower resolution.	Justification for proposed design is weak, and it only satisfies some constraints. The breakdown of system is missing some features and resolution is not appropriate at some places.	Proposed design is substantiated, preferably through proper analysis, and satisfies most given constraints. The breakdown of system into features seems complete, but resolution is not appropriate at some places.	Proposed design is substantiated, preferably through proper analysis, and satisfies all given constraints. The breakdown of system into features seems complete and at appropriate resolution, given students' level.
AR3	Pre-lab submission	Pre-lab is not submitted.	-	-	Timely submission of pre-lab and at the start of the lab time.
AR4	*Report Submission	Not accepted after 1 week.	Late submission after 2 days and within 1 week.	Late submission after the lab timing and within 2 days of the due date.	Timely submission of the report and in the lab time.